TL 101 — Assignment/Practice Sheet 2: Autodesk Fusion Certification Track

At a glance

Complete all six Autodesk Fusion courses, earn all six certificates, and submit them with your practice portfolio between **1 and 7 November 2026**.

Item	Details
Type	Individual, long-term practice assignment
Released	Monday, 5 October 2026
Submission window	Sunday, 1 November to Saturday, 7 November 2026 (hard deadline: 7 Nov, 11:59 pm)
Platform	Autodesk Learn (free, self-paced, online)
Software	Autodesk Fusion with a free Education licence
Deliverables	4 certificates of completion + 4 practice tasks, combined into one portfolio PDF

Why this matters

Every product you build in TL 101 starts as a CAD model and ends as a print on the lab's 3D printers. This track takes you through that full chain in Autodesk Fusion: model the part, document it, optimise it, plan how it gets made, and prepare it for printing.

The courses are hands-on. You learn by building alongside each lesson, so treat them as practice sessions, not videos to watch. Students who finish this track will move much faster when team product development begins.

The four courses, in order

Take the courses in this sequence; each one builds on the skills from the one before.

#	Course	What you will be able to do
1	Learn Fusion for CAD in 90 minutes	Sketch, extrude and assemble parts into a 3D model
2	Drawing	Turn a 3D model into a dimensioned 2D engineering drawing

#	Course	What you will be able to do
3	Generative design	Set loads and constraints and let Fusion propose lighter, stronger shapes
4	CAM additive manufacturing	Orient, support and prepare a part for 3D printing

Suggested pacing

At roughly one to two courses a week, you finish all five by 1 November and have the full submission week as a buffer.

Practice tasks

After each course, apply what you learnt to one part of your own. Use the same part across tasks 1 to 3 and task 6, so your portfolio shows one product moving from idea to printable file.

1. **CAD:** Model a useful object of your own design (a phone stand, cable organiser, desk hook or a part for your team idea). Minimum: one sketch-based feature, one fillet or chamfer, and real dimensions in mm. Submit: 2 screenshots (isometric and one other view).
2. **Drawing:** Create a 2D drawing of your Task 1 part with three orthographic views, an isometric view, key dimensions and a title block with your name and roll number. Submit: drawing exported as PDF.
3. **Generative design:** Run a generative study on a simple bracket or on a load-bearing feature of your part. Define the load, fixed faces and material. Submit: screenshot of 2 outcomes and 3 to 4 lines on which you would choose and why.
4. **Additive:** Prepare your Task 1 part for printing: choose orientation, add supports if needed, and note the estimated print time and material. Export an STL or 3MF that could go to the lab's K1 Max printers. Submit: screenshot of the print setup, the exported file, and 2 lines explaining your orientation choice.

How to submit

Submit one portfolio PDF and one ZIP between 1 and 7 November 2026.

- ☐ **Portfolio PDF**, named `RollNo_Name_TL101_A2.pdf`: a cover page (name, roll number, batch), then one page per course with its certificate followed by its practice task output
- ☐ **ZIP file**, named `RollNo_TL101_A2.zip`: your Fusion design files (.f3d) and the STL/3MF from Task 4
- ☐ Upload both to the google form (link will be shared on 1st Nov)

Certificates must show your own name. Download each one as a PDF as soon as you finish the course, so you are not hunting for them in November.

Evaluation and rules

Certificates prove you finished; practice tasks prove you can apply it. Both count.

Component	Weight
4 certificates of completion	50%
4 practice tasks (correctness, effort, clarity)	40%
Portfolio presentation and file naming	10%

- This is an individual assignment. Discuss freely, but every model, drawing and file must be your own work.
- Copied models or screenshots from another student get zero for that task, for both students.
- Late submissions after 7 November will not be accepted, except on documented medical grounds.
- TAs may ask you to open your .f3d file and explain a feature during a lab session.

Tips and getting help

- Keep Fusion open beside the course and build along with every lesson; pausing to try is faster than rewatching.
- One course a week, one sitting at a time, beats six courses in the last weekend.
- Use metric units (mm) throughout, to match the lab's machines.
- Stuck on a step? Check the course's own resources first, then bring your file to the next Tinkerers' Lab session and ask a TA.
- Finished early? Print your Task 4 part on the K1 Max during open lab hours and add a photo to your portfolio.